4.1

Extreme Values

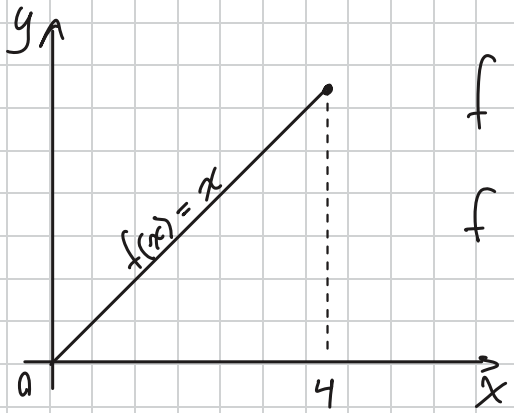
f function defined on an interval I

def 1 f has an <u>absolute maximum</u> at $c \in I$ if $f(x) \leq f(c)$ for any $x \in I$

def 2 f has an <u>absolute minimum</u> at $c \in I$ if $f(x) \geq f(c)$ for any $x \in I$

theorem f continuous on a closed, finite interval $[a, b]$ => f will have an absolute min and absolute max on I

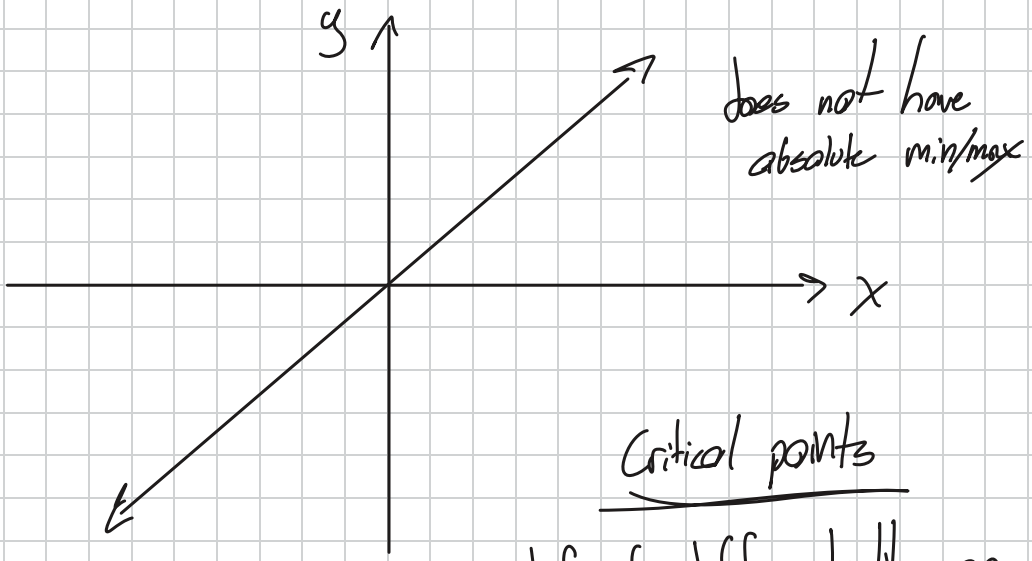
Ex. $f(x) = x$ on $I = [0, 4]$



f has an absolute min at $x = 0$

f has an absolute max at $x = 4$

Ex $f(x) = x$, $I = \mathbb{R}$



does not have absolute min/max

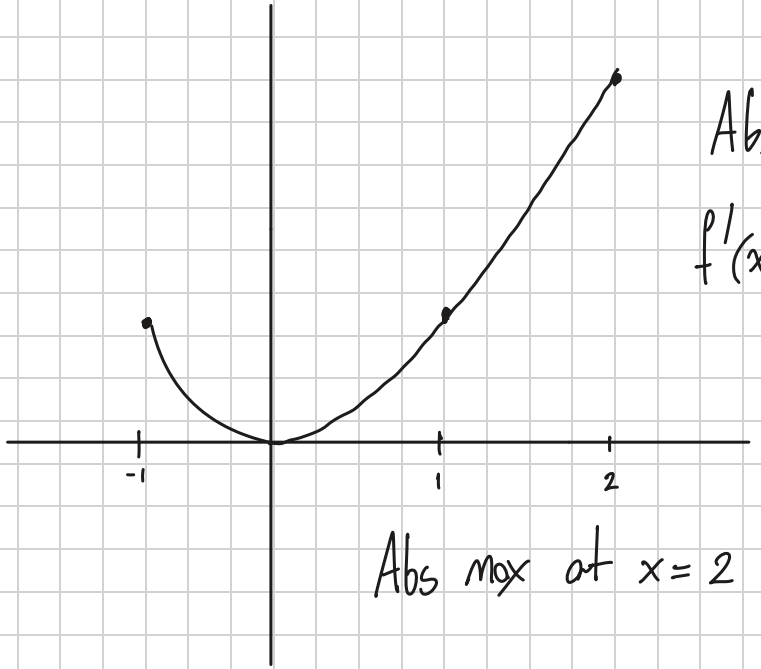
Critical points

def f differentiable on I

$c \in I$ is a critical point of f if $f'(c) = 0$

Def 4 $c \in I$ is a Singular point if $f'(c)$ DNE

Example $f(x) = x^2$, $I = [-1, 2]$

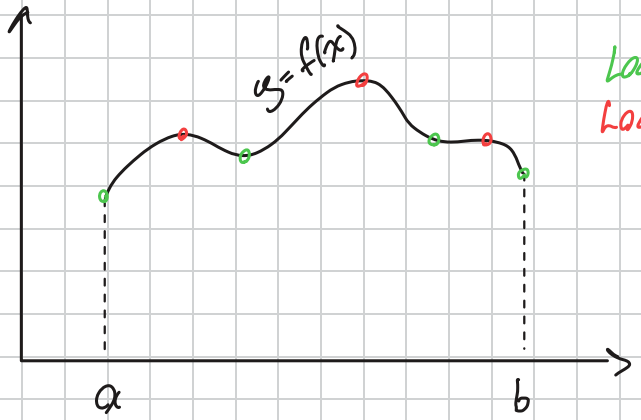


Abs min at $x = 0$

$f'(x) = 2x = 0$

$\therefore$ this is a critical point

Abs max at $x = 2$

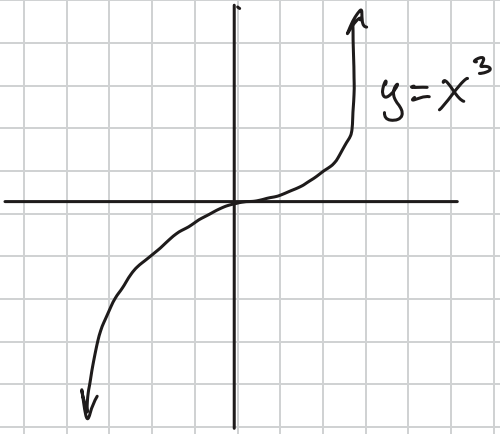


Local minimums
Local maximums

Ex

$f(x) = x^3$ $f'(x) = 3x^2 = 0$

$x = 0$
Critical point



not a local extreme value

testing

f ± n ±

determin if its a max or min value of $f'(x) = 0$